Experimental, and Conceptual Problems, Ph. Blanchard, D. Giulini, E. Joos, C. Kiefer, I.-O. Stamatescu, eds. [Berlin: Springer, 2000]).

Chapter 8

1. To be more precise, the symmetry between the laws in Connecticut and the laws in New York makes use of both translational symmetry *and* rotational symmetry. When you perform in New York, not only will you have changed location from Connecticut, but more than likely you will undertake your routines while facing in a somewhat different direction (east versus north, perhaps) than during practice.

2. Newton's laws of motion are usually described as being relevant for "inertial observers," but when one looks closely at how such observers are specified, it sounds circular: inertial observers are those observers for whom Newton's laws hold. A good way to think about what's really going on is that Newton's laws draw our attention to a large and particularly useful class of observers: those whose description of motion fits completely and quantitatively within Newton's framework. By definition, these are inertial observers. Operationally, inertial observers are those on whom no forces of any kind are acting— observers, that is, who experience no accelerations. Einstein's general relativity, by contrast, applies to all observers, regardless of their state of motion.

3. If we lived in an era during which *all* change stopped, we'd experience no passage of time (all body and brain functions would be frozen as well). But whether this would mean that the spacetime block in Figure 5.1 came to an end, or, instead, carried on with no change along the time axis—that is, whether time would come to an end or would still exist in some kind of formal, overarching sense—is a hypothetical question that's both difficult to answer and largely irrelevant for anything we might measure or experience. Note that this hypothetical situation is different from a state of maximal disorder in which entropy can't further increase, but microscopic change, like gas molecules going this way and that, still takes place.

4. The cosmic microwave radiation was discovered in 1964 by the Bell Laboratory scientists Arno Penzias and Robert Wilson, while testing a large antenna intended for use in satellite communications. Penzias and